

Factor it. Then solve it.

every step below is the same problem, one move further along

$$12x^2 - 22x - 20 = 0$$

1 Is there a GCF? Take it out.

every term is even — so divide the whole equation by 2

ALL THREE SHARE A 2

$$12x^2 - 22x - 20 = 0$$

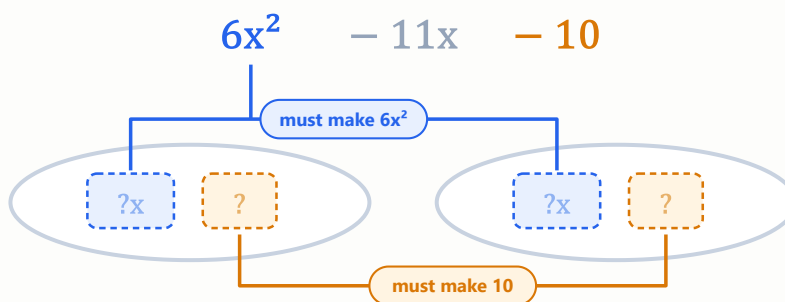
 $\div 2$

THE 2 IS GONE — IT NEVER COMES BACK

$$6x^2 - 11x - 10 = 0$$

2 Three terms always make two bubbles.

now decide what has to go where

**3 List every way to make each one.**

then start with the pair that is closest together

WAYS TO MAKE 6

$6 \cdot 1$

$3 \cdot 2$

closest pair —
always start here

WAYS TO MAKE 10

$10 \cdot 1$

$2 \cdot 5$

closest pair —
always start here**4 Drop the numbers in. Leave the signs out.**

right now we are only testing positions

**Never leave a GCF inside a bubble.**

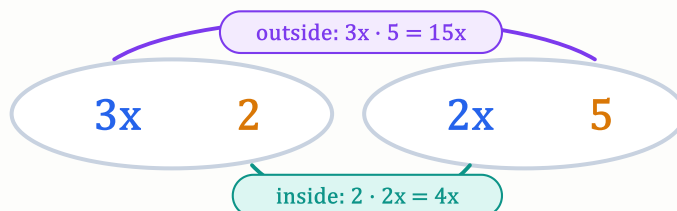
You already divided the 2 out — so a pair like this can't happen:

$2x \ 10$

both share a 2

5 Multiply the inside pair and the outside pair.

together they have to make the middle number, 11



CAN 15 AND 4 MAKE 11?

$$15 - 4 = 11$$

✓ yes — these are the right spots

If they can't make 11 → swap the 2 and the 5. Still can't → go back and swap the 3x and the 2x.

6 Now the signs place themselves.the middle term is -11 , so one of them has to turn negativewe need -11 , not $+11$

$$-15 + 4 = -11$$

the 15 came from $3x \cdot 5$

so the 5 goes negative



$$(3x + 2)(2x - 5)$$

check the last spots: $(+2)(-5) = -10$ ✓**7 Set each bubble to 0.**

two things multiply to zero → one of them must be zero

$$(3x + 2)(2x - 5) = 0$$

$$3x + 2 = 0$$

$$x = -\frac{2}{3}$$

$$2x - 5 = 0$$

$$x = \frac{5}{2}$$